

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 13-101-2
DEADBREAK ELBOWS (SCREENED SEPARABLE INSULATED CONNECTORS) FOR 12/20(24) KV M.V XLPE INSULATED CABLES	3-02-2026

EDMS 13-101-2

SPECIFICATION

FOR

DEADBREAK ELBOWS

(SCREENED SEPARABLE INSULATED CONNECTORS)

FOR 12/20(24) kV MV XLPE INSULATED CABLE

Issue: Feb. 2026/ Rev-2

- This revision contains option items that should be selected by EDC before bidding.

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1. SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacture, testing, inspection, installation and performance of Dead break Elbows (screened separable insulated connectors) for 12 KV system (12/20kv cables). The elbows can be L or T Shape. The elbows bushing is Outside Cone Interfaces type C1&C2 up to 1250 A bolted screw type, and conforms to EN 50181. but the tender shall supply elbow for other interface types according to EDC requirements.

Elbows will be used with power cables, XLPE insulation, with the different construction:

- I. Three core and single core.
- II. Armored and non-armored.
- III. Copper and AL conductors.
- IV. Conductor range up to 630 mm².
- V. The materials of the elbow shall be compatible with the materials of the cables.

2. DEFINITIONS

2.1. Conductor Temperature:

The termination should be suitable for use on cable shaving maximum conductor temperature of 90°C at normal conditions, and maximum temperature of the cable conductors under short-circuit condition of 250°C.

2.2. Rated voltage: U₀/U (U_m)

The rated voltages U₀/U(U_m) of the cable termination considered in this specification is 12/20 (24) Kv.

- U₀: is the rated power frequency voltage between conductor and earth or metallic screen for which the cable is designed (phase voltage).
- U: is the rated power frequency voltage between conductors for which the cable is designed (line voltage);
- U_m: is the maximum value of the "highest system voltage" for which the equipment may be used.

3. APPLICABLE STANDARDS

The elbow Terminations covered in this specification shall comply with the latest edition/amendment of the following standards family IEC or CENELEC given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall

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apply. Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.

S. N.o	Cross Reference	Description
1.	EDMS 15-300	COMPRESSION TYPE LUGS FOR CABLES UP TO 24 kV

Table no. 1

<u>S No.</u>	<u>Standard No.</u>	<u>Description</u>
2.	IEC 60502-2	Cables for rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
3.	IEC 61238-1-3	Compression and mechanical connectors for power cables - Part Test methods and requirements for compression and mechanical connectors for power cables for rated voltages above 1 kV ($U_m = 1,2$ kV) up to 30 kV ($U_m = 36$ kV) tested on non-insulated conductors
4.	IEC 60502-4	Test requirements on accessories for cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
5.	IEC 61442	Test methods for accessories for power cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
6.	IEC 60228	Conductors of insulated cables
7.	CENELEC HD 629.1S3	Test requirements on accessories for use on power cables rated voltage from 3.6/6(7.2) kV up to 20.8/36(42) Kv - Part 1
8.	IEC 60060	High-voltage test techniques
9.	IEC 60230	Impulse tests on cables and their accessories
10.	IEC 60885-3	Electrical test methods for electric cables Part 3: Test methods for partial discharge measurements on lengths of extruded power cables
11.	EN50180	Bushings above 1kV up to 52 kV and from 250 A to 3.15 kA for liquid filled transformers
12.	EN50181	Plug-in type bushings above 1 kV up to 52 kV and from 250 A to 2.50 kA for equipment other than liquid filled transformers
13.	IEC 60137	Insulated bushings for alternating voltages above 1000 V
14.	IEEE 386	Standard for Separable Insulated Connector Systems for Power Distribution Systems above 600 V

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4. ENVIRONMENTAL CONDITIONS

The performance of the Termination components in the kits should be guaranteed for following the environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements),
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

5. SYSTEM DETAILS:

Frequency:	50 HZ.
Number of phases:	One or Three.
Nominal voltage:	11 KV.
Highest voltage:	12 KV.
Maximum prospective Fault level:	25 KA. For Duration of 1 sec.

6. SEPARABLE CONNECTOR DESIGN, RATINGS, AND PERFORMANCE:

- 6.1. The separable connector housing (elbow) shall be factory molded in the shape that it will take when installed, the molded layers of the elbow housing (the inner semi conductive layer, insulation layer, and semi conductive shield layer) shall be integrated during the molding injection processes, and made of ethylene propylene diene monomer (EPDM) (or other material with equivalent performance) .
- 6.2. Elbow housing and cable adaptor have integrated geometrical stress control, using of tapes or mastic paste for control/damping the electrical fields are **not** accepted.
- 6.3. Elbow housing insulation thickness shall not be less than the cable insulation thickness.
- 6.4. Elbow housing has semi conductive integrated faraday cage surrounding the elbow lug and connections area. The minimum thickness for this layer 3.0 mm.
- 6.5. The minimum thickness at any point of the outer semi conductive layer is 2.5 mm (excluding silicon-rubber type), painted outer semi conductive layer is **not** accepted, the maximum surface resistance of elbow housing is 5k Ω ,
- 6.6. The insulation thickness of the cable adaptor shall not less than 6 mm (unstretched - as applicable).
- 6.7. 100% of the elbow housing Production shall be electrically testing in the factory.

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- 6.8. The Installation is performed by sliding the elbow housing and adaptor over a prepared cable and connected the cable lugs with the interfacing bushing by connecting stud.
- 6.9. **The use of heat is not a part of the installation procedure except the breakout and outer jacket tubes.**
- 6.10. The outer semi conductive layer of the elbow housing shall be over lapped with cable adaptor semi conductive layer by at least 10 mm.
- 6.11. The supplier shall ensure that the preparation techniques of the medium-voltage elbow kits are adhering to accepted best industry practices, and international guidelines.
- 6.12. The cross-sectional area of the copper Braid shall be at least equal to the effective cross-sectional area of the cable copper screen with minimum size as indicated in the following table.

Minimum cooper screen of elbow for single and three core cables 12/20 kv

C.S.A (mm2)	35	50	70	95	120	150	185	240	300	400	500	630
Total Copper weight for the kit gm/m	130					200				315		
Min. length of the copper Baird per kit cm	90											

- 6.13. Each kit should be supplied with installation instructions in Arabic (and English).
- 6.14. **The vendor shall be deliver any component of elbow as required of –EDC.**
- 6.15. **ADDIONAL ACCESSORIES FOR TERMINATION KIT**
- 6.17.1. Cleaning solvent >100 c.(m.l) + cleaning tissues (3 tissues per phase), or cleaning wipes as the following:
- 12 wipes for 3 core / 6 for single core termination.
 - Wipe: fiber (20*20cm)
 - Solvent type: isopropyl or tetrachloroethylene
- 6.17.2. Silicon grease: (9 Packets) each one > 5 gm or (6 Packets) each one > 10 gm (for 3 core cables).
- 6.17.3. Galvanized steel clamps: 1 pcs for binding steel armour layer. (only applicable For Three Core cables).
- 6.17.4. Stainless clamp: 1 Pcs for binding AL armour layer (only for single core).
- 6.17.5. Mastick paste (or tape) to be used as sealant at least at:
- Between breakout & cable outer jacket.

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- Underneath and over cu. braid, cable steel tape and cu. tape to cover welding point.

6.17.6. 3 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut).(for 3 cores).

6.17.7. 1 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut).(for 1 core).

6.17.8. 1 piece of sandpaper size 320 with length 40 cm, cloth type (non-easy cut).

7. ROUTINE TESTS :

- The tests made by the manufacturer on each finished product or components of accessories in the manufacturer facility to check that each product/ component meets the specified requirements.

System Voltage	Partial discharge (corona) voltage test discharge magnitude (<10 pc) The values of tests as per IEC 60502-4	AC withstand voltage (5 minute) The values of tests as per IEC 60502-2
12 KV	20 KV	48 KV

- Vendor should keep records for the routine tests
- Vendor should keep records for the calibration of the instruments and testing facilities which using in the routine tests

8. SAMPLE AND INSPECTION TEST

- 8.1. This clause is applicable only for inspection by –EDC on order before delivery and not be a substitute of the routine test which require to be done on 100% of the order quantity by manufacturer as described in cluse 7
- 8.2. Sample tests should be made by the manufacturer on samples of finished product or components taken from a finished product store, at a specified frequency, so as to verify that the finished product meets the specified requirements.
- 8.3. Inspection test should be done in the manufacturer/vendor facilities in attendance ofEDC representatives per each order before delivery..
- 8.4. Inspection tests should be conducted as per submitted/approved inspection plan during the tender stage,
- 8.5. the inspection plan should include a visual examination on each component, there should be no visual defects in the material, manufacture, workmanship and markings.
- 8.6. The Tenderer should provide free of charge all the testing facilities required at work.

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8.7. The inspection sample should be 1% of the order quantity and should not be more than 5 pcs of each elbow size.

9. TYPE TESTS:

- 9.1. Type Testing should be made to demonstrate satisfactory performance characteristics according to the international standard to meet the intended application.
- 9.2. It is not necessary to repeat the tests, once successfully completed, unless changes are made in the materials, design or manufacturing process which might affect the performance characteristics
- 9.3. Cables used for testing shall comply with IEC60502-2 and shall be of the same rated voltage.
- 9.4. Accessories should be assembled in the manner specified by the manufacturer's instructions.
- 9.5. Type tests should be done in independent and approved testing laboratory
- 9.6. Type test for cable elbow should be conducted as per IEC60502-4 table12 or CENELEC HD 629-1S3 standards table 14.
- 9.7. Test method shall be as per IEC 61442 or CENELC HD 629.1S3 standards.
- 9.8. The summary test voltage as per IEC60502-4 table17.
- 9.9. The test program dates should be advised by the contractor in advance to get preapproval on the type test plan.
- 9.10. Tests described above should not release contract or from his full responsibility towards supplies, guarantee, and compliance with this specification.
- 9.11. Test cable conductor cross-sections for separable connector shall be according to (IEC 60502-4 table3).
- 9.12. To extend the approval range of the type test for smaller or larger cross section area, additional tests as per (IEC 60502-4 table 14&15) or (CENELEC HD 629-1S3 table 17&18) shall be applied on the appropriate larger or smaller cross-section, but the extension of compliance of separable connectors to larger cable conductor cross sections, the test current shall be limited to the rating current of the bushing, and cable adaptor size.
- 9.13. Type test done on higher voltage class shall cover all the same or lower voltage classes, example type test done for voltage class 36 kv will cover all voltage classes 12 kv, 24 kv , and 36 kv.
- 9.14. No. of samples of elbow for type testing the 3core cable should be 6 samples, each two samples installed on one phase of the cable connected back-to-back by a connector plug or bushing, the below diagram no.1 just for clarification.
- 9.15. Tests no. 13,14 and 19 in the table 12 are applicable only for elbow interface A or B, and are not applicable for elbow interface C and D.

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IEC 60502-4 Table 12 – Test sequences and requirements for screened dead-break separable connectors

Tests ^a		Requirements	Test methods of IEC 61442	Test sequences (see Figure 4)			
				4.1	4.2	4.3	4.4
1	AC voltage dry	5 min at 4,5 U_0 . No breakdown.	Clause 5	X	X	X	
2	Partial discharge at ambient	10 pC max. at 1,73 U_0^b .	Clause 8	X			
3	Impulse at θ_t	10 impulses of each polarity. No breakdown.	Clause 7	X			
4	Thermal short-circuit (screen) ^c	Two short-circuits at I_{SC} of the cable screen. No visible deterioration.	Clause 11		X	X	
5	Thermal short-circuit (conductor)	Two short-circuits to raise conductor to θ_{SC} of the cable. No visible deterioration.	Clause 12		X	X ^d	
6	Dynamic short-circuit for three-core cable only ^{e d}	One short-circuit at I_d . No visible deterioration.	Clause 13			X ^d	
7	Heating cycles in air	Min. 30 cycles at θ_t and 2,5 U_0^1 .	Clause 10	X			
8	Heating cycles in water	Min. 30 cycles at θ_t and 2,5 U_0^1 .	Clause 10	X			
9	Disconnect/connect ^f	Five times. (No visible damage to contact.)	–	X	X	X	
10	Partial discharge at θ_t and ambient temperature	10 pC max. at 1,73 U_0^b .	Clause 8	X			
11	Impulse at ambient	10 impulses of each polarity. No breakdown.	Clause 7	X	X	X	
12	AC voltage dry	5 min at 4,5 U_0 . No breakdown.	Clause 5	X	X	X	
13	Operating eye	Axial force 1 300 N for 1 min. Torque 14 N · m.	Clause 20				X
14	Partial discharge at ambient	10 pC max. at 1,73 U_0^b .	Clause 8				X
15	Examination according to Annex C	Shall be documented in the report.	–	X	X	X	X
16	Screen resistance ^g	Maximum 5 000 Ω .	Clause 16	Tests 16 to 20 are carried out on separate samples. For tests 16 and 19, no cable is required. For tests 17, 18 and 20, appropriate cable lengths shall be used.			
17	Screen leakage current ^g	Maximum 0,5 mA at U_m .	Clause 17				
18	Fault current initiation	See notes ^{g h} .	Clause 18				
19	Operating force	Force < 900 N.	Clause 19				
20	Capacitive test point	Capacitance of test point to cable conductor: $C_{tc} > 1,0$ pF. Ratio of capacitance of test point to earth C_{te} and capacitance of test point to cable conductor C_{tc} :	Clause 21				

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$$C_{te}/C_{tc} \leq 12,0.$$

- The tests shall be carried out at ambient temperature, unless otherwise specified.
- Not required for accessories installed on 3,6/6 (7,2) kV cables having unscreened insulation. This requirement can also be achieved with partial discharge tests conducted at 2 U₀.
- This test applies only to separable connectors that are equipped with a connection to, or adaptor for, the metallic screen of the cable.
- The thermal short-circuit may be combined with the dynamic short-circuit.
- Only required for single-core cable accessories designed for initial peak currents $I_p > 80$ kA and three-core accessories designed for $I_p > 63$ kA. For the value of the current I_d see 9.2.
- The test shall be carried out only when the cable is de-energized.
- The test is required for separable connectors without a metallic housing or with a removable metallic housing. The metallic housing shall be removed prior to the test. This test is not required for separable connectors which can only be used in service with the metallic housing in position.
- For solidly earthed systems, the fault initiation shall occur within 3 s. For unearthed or impedance earthed systems, the fault current shall flow continuously.
- Current, see Table 3.

HD 629.1 S3 Table 14 — Screened separable connectors for extruded insulation cables

Test	Test clause of EN 61442	Test sequence			Test requirements
		D1	D2	D3 (a)	
1 AC voltage withstand dry	4	x	x		5 min at 4,5 U ₀ , no breakdown
2 Partial discharge at ambient temperature	7	x			max.10 pC at 2 U ₀
3 Impulse voltage at elevated temperature	6	x			10 impulses of each polarity, no breakdown
4 Heating cycle voltage in air	9	x			63 cycles at 2,5 U ₀ , no breakdown
5 Heating cycle voltage in water	9	x			63 cycles at 2,5 U ₀ , no breakdown
6 Thermal short circuit (screen) ^(b)	10		x		2 short circuits at I _{sc} , no breakdown
7 Thermal short circuit (conductor)	11		x		2 short circuits to raise conductor to θ_{sc} of the cable, no breakdown
8 Dynamic short circuit for 3-core cable only ^(c)	12		x		1 short circuit at I _d ^(d) , no breakdown
9 Disconnection/connection ^(e)	-	x	x		5 complete operations, no visible damage to contact
10 Partial discharge at elevated and ambient temperature	7	x			max. 10 pC at 2 U ₀
11 Impulse voltage at ambient temperature	6	x	x		10 impulses of each polarity, no breakdown
12 AC voltage withstand dry	4	x	x		5 min at 4,5 U ₀ , no breakdown

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13	Operating eye	19			x	Axial force: 1 300 N for 1 minTorque: 14 Nm No visible damage
14	Partial discharge at ambienttemperature	7	x		x	max. 10 pC at 2 U_0
15	Visual examination according to Annex C	-	x	x	x	Shall be documented in the report
16	Screen resistance measurement ^(f)	15	See (f)			max. 5 000 Ω
17	Leakage current measurement ^(f)	16				max. 0,5 mA at U_m
18	Screen fault current initiation	17				Solidly earthed system: - fault initiation to occur within 3 s Unearthed/impedance earthed system: - fault current to flow continuously
19	Operating force	18				Force less than 900 N
20	Capacitive test point performance ^(h)	20				Capacitance of test point to cable conductor $C_{tc} > 1,0$ pF Ratio of capacitance of test point to earth C_{te} and capacitance of test point to cable conductor C_{tc} : $C_{te}/C_{tc} \leq 12$

- (a) If fitted with operating eye.
- (b) This test is only required for accessories that are equipped with a connection to, or adaptor for, the metallic screen of the cable.
- (c) This test can be combined with the thermal short circuit on the conductor.
- (d) For the value of the current I_d , see 7.2.
- (e) The test shall be carried out only when the cable is de-energized.
- (f) The test is required for separable connectors without a metallic housing or with a removable metallic housing. The metallic housing shall be removed prior to the test.
This test is not required for separable connectors which can only be used in service with the metallic housing in position.
- (g) Tests 16 to 20 are carried out on separate samples.
For tests 16 and 19, no cable is required.
For tests 17, 18 and 20 appropriate cable lengths shall be used.
- (h) If fitted with capacitive test point.

NOTE Refer to Clause 8 of EN 61442 for details of the calibration of conductor heating current required to have been carried out before the commencement of these tests.

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IEC 60502-4 Table 14 – Additional tests for smaller conductor cross-sectional areas

Tests ^a		Requirements	Test methods of IEC 61442	Test sequences (see Figure 1, Figure 2, Figure 3)		
				1.1 ^b	2.1 ^c	3.1 ^d
1	AC voltage dry	5 min at 4,5 U_0 . No breakdown nor flashover.	Clause 5	X	X	X
2	Partial discharge at ambient	10 pC max. at 1,73 U_0 ^e	Clause 8	X	X	X
3	Impulse	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X	X	X
4	Heating cycles	No breakdown no flashover. Screened and unscreened separable connectors. Min. 10 cycles ^f in water at θ_t ^g , and 2,5 U_0 .	10.2 and 10.3	X	X	X
5	AC voltage dry	5 min at 4,5 U_0 . No breakdown nor flashover.	Clause 5	X	X	X
6	Partial discharge at θ_t ^g , and ambient temperature	10 pC max. at 1,73 U_0 ^e .	Clause 8	X	X	X
7	Examination according to Annex C	Shall be documented in the report.	–	X	X	X
^a The tests shall be carried out at ambient temperature, unless otherwise specified. ^e Not required for accessories installed on 3,6/6 (7,2) kV cables having unscreened insulation. This requirement can also be achieved with partial discharge tests conducted at 2 U_0 . ^g θ_t is the maximum cable conductor temperature in normal operation +5 K to 10 K.						

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HD 629.1 S3 Table 17 — Additional tests for smallest cable cross section compliance ⁽¹⁾

Test	Test clause of EN 61442	Test sequence	Test requirements
1 AC voltage withstand dry	4	x	5 min at 4,5 U_0 , no breakdown norflashover
2 Partial discharge at ambient temperature	7	x	max. 10 pC at 2 U_0
3 Impulse at ambient temperature	6	x	10 impulses of each polarity, no breakdown nor flashover
4 Heating cycle voltage in air or water	9	x	Joints, Screened and Unscreened Separable connectors 12 cycles in water at 2,5 U_0 ,
5 AC voltage withstand dry	4	x	5 min at 4,5 U_0 , no breakdown norflashover
6 Partial discharge at elevated and ambient temperature	7	x	max. 10 pC at 2 U_0
7 Visual examination according to Annex C	-	x	Shall be documented in the report
(1) Number of samples tested Single core: 2 samples, 3-core 1 sample – Screened separable connectors : see Figure 7			
NOTE Refer to Clause 8 of EN 61442 for details of the calibration of conductor heating current required to have been carried out before the commencement of these tests.			

IEC 60502-4 Table 15 – Additional tests for separable connector compliance extension to largest cable cross-section

Tests ^{a, c}		Requirements	Test methods of IEC 61442	Test sequences (see Figure 4 and Figure 5)		
1	AC voltage dry	5 min at 4,5 U_0 . No breakdown norflashover.	Clause 5	X		
2	Impulse at ambient	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X		
3	AC voltage dry	5 min at 4,5 U_0 . No breakdown norflashover.	Clause 5	X		
4	Partial discharge at ambient temperature	10 pC max. at 1,73 U_0 ^b .	Clause 8	X		
5	Examination according to Annex C	Shall be documented in the report.	-	X		

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- a The tests shall be carried out at ambient temperature, unless otherwise specified.
- b This requirement can also be achieved with partial discharge tests conducted at $2 U_0$.
- c The number of samples tested is as follows:
- screened separable connectors: see Figure 4.

HD629.1 S3 Table 18 — Additional tests for separable connector compliance extension to largest cable cross section ⁽¹⁾

Test	Test clause of EN 61442	Test sequence	Test requirements
1 AC voltage withstand dry	4	x	5 min at 4,5 U_0 , no breakdown nor flashover
2 Impulse at ambient temperature	6	x	10 impulses of each polarity, nobreakdown nor flashover
3 AC voltage withstand dry	4	x	5 min at 4,5 U_0 , no breakdown nor flashover
4 Partial discharge at ambient temperature	7	x	max. 10 pC at 2 U_0
Visual examination according to Annex C	-	x	Shall be documented in the report
(1) Number of samples tested: – Screened separable connectors:		see Figure 7,	

IEC 60502-4 Table 16 – Summary of tests

Tests	Separable connectors	Tests	Separable connectors
	Screened Dead-break		Screened Deadbreak
AC voltage – 4,5 U_0 /5 min dry	x		
Partial discharge – at θ_t & at ambient temperature	x	Screen resistance	x
Impulse – at θ_t & at ambient temperature	x	Screen leakage current	x
Heating cycles – in air & in water	x	Fault current initiation	x
Thermal short-circuit			

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– screen & conductor	x	Capacitive test point	x
Dynamic short-circuit	x		
Disconnect/connect	x	Examination	x

IEC 60502-4 Table 17 – Summary of test voltages (see Clause 9)

Rated voltage U_0/U (U_m) KV						
Test	Test voltage	3,6/6 (7,2)	6/10 (12)	8,7/15 (17,5)	12/20 (24)	18/30 (36)
Humidity and salt fog	1,25 U_0	4,5	7,5	11	15	22,5
Partial discharge	1,73 U_0	6	10	15	20	30
Heating cycle voltage and AC voltage/500 h	2,5 U_0	9	15	22	30	45
AC voltage/1 min	4 U_0	14,5	24	35	48	72
AC voltage/5 min	4,5 U_0	16	27	39	54	81
Impulse (peak)	—	60	75	95	125	170
^a Not required for accessories installed on 3,6/6 (7,2) kV cables having unscreened insulation. This requirement can also be achieved with partial discharge tests conducted at 2 U_0 .						

HD629.1 S3 Table 19 – Summary of test voltages

Test	Test voltage	Rated voltage U_0/U (U_m) kV									
		3,6/6 (7,2)	3,8/6,6 (7,2)	6/10 (12)	6,35/11 (12)	8,7/15 (17,5)	12/20 (24)	12,7/22 (24)	18/30 (36)	19/33 (36)	20,8/36 (42)
Humidity and salt fog	1,25 U_0	4,5	5,0	7,5	8	11	15	16	22,5	24	26
Partial discharge	2 U_0	7	7,5	12	12,5	17,5	24	25	36	38	42
Heating cycle voltage and AC voltage / 500 h	2,5 U_0	9	9,5	15	16	23	30	32	45	47,5	52
AC voltage withstand / 1 min	4 U_0	14,5	15	24	25,5	35	48	51	72	76	83
AC voltage withstand / 5 min	4,5 U_0	16	17	27	28,5	39	54	57	81	85,5	93,5
Impulse (peak)	-	60	60	75	95	95	125	125	170	194	200

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IEC 60502-4 Table 3 – Test cable conductor cross-sections for separable connectors

Rating of separable connector (A)	Cable conductor cross-section (mm ²)			
	Test sequence 4.1/5.1 ^a		Test sequence 4.2/5.2 ^{c, d}	
	Cu	Al	Cu	Al
250	50	70	50	70
400	95	150	95	120
630	185	300	120	185
800	300	400	150	240
1 250	500	630	240	400
2 500 ^b	2 × 500	2 × 630	500	800

^a The use of the cross-sections in this Table 3 can lead to overheating of the bushing while achieving the required conductor temperature. Under these circumstances, it is permissible to use a conductor one size smaller. If bushing failure occurs, the test will be declared void (see 10.3).

^b MV cables with 2 500 A rating are not readily available. Separable connectors for this rating are therefore tested with a dual cable connection, where two cables connected in parallel each supply half of the total current.

^c The equipment bushing short-circuit rating of $25 \times I_r$ (IEC 60137:2017, 4.3 [6]) is limiting the short-circuit current value. Cross-sections listed are selected to reach the cable short-circuit temperature θ_{SC} .

^d Test sequence 4.2/5.2 can be done on a cable cross-section from test sequence 4.1/5.1.

10. MARKING:

The components of the Separable connector kit should be distinctly and clearly marked with the manufacturer's name.

All components should be capable of being stored without deterioration within the temperature range of -5°C to +45°C (+50 °C as option). Components or materials, if subjected to a shelf-life limitation, should have the final date of use prominently and permanently shown on all packages.

The elbow housing (main insulation part) must be marked with the following data:-

1. Manufacturer name.
2. Production indicator for traceability.
3. Cross-section, diameter, or application range should be marked.

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11. SUBMITTALS

The tenderer should submit with his offer a duly completed and signed package of all the required documents as:

- 11.1 A reference list showing that the proposed technique has been extensively used by recognized utility companies outside the manufacturer's country for 3 years (for foreign manufacturers only).
- 11.2 Copies of recently (latest) conducted type test certificates (electrical and mechanical) or qualification reports from certified independent and approved test laboratory for the offered terminations that verify all the requirements as listed at table (12) of IEC 60502-4 or VDE 0278-629-1 S3 or all the required as listed at table (14) of CENELEC HD 629-1 S3 standards.
- 11.3 Data sheet for the lugs used within the accessories should include at least the following data:
 - 11.3.1 Lug material details (alloy, conductivity, chemical composition ...)
 - 11.3.2 Assembly technique,
 - 11.3.3 Tooling, dies and necessary setting,
 - 11.3.4 Preparation of contact surfaces, if applicable,
 - 11.3.5 Type, reference number and any other identification of the lug,
- 11.4 The Factory Acceptance Test (FAT), which carried out at manufacturer/vendor facilities on each finished product and components of accessories to check that each product and components meets the specified requirements
- 11.5 A detailed list of inspection tests that have been carried out at manufacturer's facilities in attendance of the ...EDC representative(s).
- 11.6 The name of the laboratory where tests can be conducted
- 11.7 Complete technical details with dimensional drawings, showing description of material, the following data at least should be included:
 - 11.7.1. Name of manufacturer and manufacturing location.
 - 11.7.2. Type, designation, or code.
 - 11.7.3. Rated voltage,
 - 11.7.4. Minimum and maximum cable insulation diameters,
 - 11.7.5. Minimum and maximum cable cross-sections for this rated voltage
 - 11.7.6. Applicable cable construction including the material and shape of cable conductor
 - 11.7.7. Installation instructions (reference)
- 11.8 Catalogue and Instructions of installation (Arabic and English preferred) of the offered accessories. The catalogues must include the number of dies, detailed drawing of the lug explaining the number of crimping and the width of each crimping, each crimping should be hexagonal.
- 11.9 Details of storage, shelf life under Egypt service conditions mentioned in clause 6 (It must not be >30 months).

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- 11.10 Lifetime of termination should not be less than 30 years starting from the date of operation.
- 11.11 One sample of complete kit will be submitted with the offer.
- 11.12 Details of any special tool requirement.
- 11.13 Schedule of guarantee duly completed and signed by the manufacturer and vendor/Agent, otherwise the tenderer should be rejected.
- 11.14 Kit content lists stating the quantities and descriptions of components.

12. PACKING:

- Packing should be robust and able to withstand very rough handling during transportation. It should be protected against ingress of moisture.
- The details of method of packing should be submitted with the offer.
- Each kit should be supplied with a packing note stating the quantities and descriptions of components contained in the kits, and the documents described below:
- Elbow Instructions of installation (Arabic and English) for the offered accessories which must include the steps of installation, the number of dies, detailed drawing of the lugs explaining the number & sequence of crimping and the width of each crimping, (Each crimping should be hexagonal).
- Packaging as well for each cable accessory should be printed with the following information:
 - 1- Manufacturer's name.
 - 2- Manufacturer's Type No.
 - 3- Cable size or cable diameter.
 - 4- Nominal Voltage.
 - 5- Manufacturing year.
 - 6- Manufacturer's batch No. or any tracing no.
 - 7- Shipping Mark Guide.

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13. KIT`S COMPONENTS:

The Kit of the ELBOW should include but not limited for the following items:

- Elbow housing “The main insulation part”.
- Cable adaptor (as applicable).
- Terminal lug.
- Insulated Plug “As applicable”.
- Insulation plug Cap “As applicable”.
- connection Stud, to connect the cable lugs with the busing interface.
- Outer jacket Heat shrinkable tube.
- Earthing connection for elbow housing and for the cable screen.

14. TRAINING:

The manufacturer / tenderer shall be responsible for training of the...EDC teams. The elbow products should be installed by proper tools as per elbow manufacturer installation instructions, records for regular checks on the used tools by the jointers should be done and recorded by ... EDC.

15. GUARANTEE:

The vendor shall guarantee supplies against all defects arising out of faulty design or workman ship or defective material for a period of two year from date of delivery to ... EDC stores, inspection and technical acceptance. the vendor shall guarantee.

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16. GUARANTEE FOR SEPARABLE CONNECTOR , PRE Molded,....X.....mm2,.....kV

Item	Description
1	Type----- Catalogue no-----
2	Name of manufacturer -----
3	Max. design voltage -----kV.
4	Period for which this accessory has been supplied for Service conditions similar to Egypt----- years
5	Average splicing time-----hrs.
	1-Compression lugs, with seal against moisture:- - Material composition (CU%)/(AL%) - No of compressions ((for compression type)).----- - No. of bolts (for share bolt type) ----- - Total length -----mm - Plate hole of lug (Diam.) -----mm - Conductor hole (lug inner diam.)----- mm 2-Adaptor:- - Number of adaptors - application range-----mm 3-Breakout (spreader cap) :-Dims in-----mm Total Length /inner diam., for cable side /inner diam. lug -Inner diam. /thickness for cable side after free shrink -----mm/mm -Inner diam./thick ness for Lug side after free shrink -
7	Any special tool required-----
8	Any special accessories required
9	Type/Quantity of cleaning solvent-----
10	Length of tin wire solder with flux-----cm.
11	Weight of silicon grease-----gm.

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12	Binding tapes: A- PVC tapes: -Length /thickness---m/mm	----- -----
13	Mastic pastes:- type/ number/ length each	----- -----
14	Any other tape: type, length-----	
15	Attach details on rated short circuit current capacity:- a) Copper mesh or Braid b) Armour c) Bonded mesh or Braid with armor	
16	Copy of works routine test reports and if requested byEDC also batch / type test reports on the kits being supplied	
17	Confirm that visual inspection and routine tests shall be carried out on each kit-----	Yes /No
18	Confirm compliance to the appropriate requirements of table2-----	Yes /No
19	Marking and packing: - a) Marking shall be to clause 4.4 ----- b) Packing shall be to clause 4.5 ----- c) Each kit with components shall be complete in itself and separately packed----- d) Submit detailed packing list	Yes /No Yes /No
20	Guarantee confirm compliance to clause 6.3----	Yes /No
21	For direct technical clarification from the manufacturer Submit the details mentioned below of the persons responsible:- Name :----- Designation :----- Address :-----	

We guarantee above mentioned data for our offered material for.....EDC.....

Vendor/ Agent.....

Important: Each tenderer shall submit with his offer the above tender" specifications for

cable.....Elbow ...x.....mm². With each page signed by the tenderer. This signature means that tenderer confirms verification of his offer for these specifications.

Tenders not accompanied with signed specifications shall not be considered.

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SCHEDULE GUARANTEE

- Schedule of deviations from-----
Technical specification
- This schedule be filled and signed by every tenderer.
- If there are no deviations or reservations. This should be explicitly stated in this schedule.

Item Number On tender	Precise details of minor deviations Included in the offer.